KAGISO KOTU

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Professional Summary

Final-year BSc Computational and Applied Mathematics student at the University of the Witwatersrand, majoring in Computer Science and Applied Mathematics. Passionate about problem solving, software development, and mathematical modelling, with strong analytical and technical skills. Experienced in developing full-stack web applications, Android applications, and intelligent software solutions through academic and collaborative projects. Comfortable working in teams and applying Agile Scrum, Test-Driven Development, and CI/CD practices.

Education

University of the Witwatersrand, Johannesburg
BSc Computational and Applied Mathematics *Final Year Student*
Majors: Computer Science and Applied Mathematics

Technical Skills

Programming	Java, JavaScript, Python, C++, SQL
Development	Object-Oriented Programming, Data Structures & Algorithms, Software Design, REST APIs, Full-Stack Development
Web	HTML, CSS, JavaScript
Databases	MySQL, SQL Database Design
Tools	Git, GitHub, GitHub Actions
Practices	Agile Scrum, Test-Driven Development, CI/CD, Software Development Lifecycle
Mathematics	Mathematical Modelling, Optimization, Applied Mathematics

Selected Projects

TNBT – Stokvel Management Platform
JavaScript, SQL, Agile Scrum, TDD, CI/CD

- Collaborated in a six-member team to develop a full-stack stokvel management web application.
- Applied Agile Scrum, Test-Driven Development, user acceptance testing, and CI/CD using GitHub Actions.
- Contributed to requirements engineering, software design, implementation, testing, and deployment.

Managrs – Freelancing Management Platform
Full-Stack Web Development

- Developed a platform for collaboration between freelancers and clients.
- Contributed to project tracking, milestone management, communication, analytics, and role-based functionality.
- Participated in software design, development, testing, and documentation.

Woofer – Social Media Android Application
Java, MySQL, PHP REST APIs

- Developed an Android social networking application with posts, comments, friend management, messaging, notifications, and profiles.
- Integrated MySQL database functionality and REST APIs for client-server communication.

Multiplayer Snake AI
Java

- Developed an intelligent game agent using pathfinding and hazard-avoidance algorithms.
- Applied Object-Oriented Programming and algorithmic problem-solving techniques.

Experience & Leadership

Applied Mathematics Tutor *University of the Witwatersrand*
Tutored first-year Applied Mathematics students, helping them develop mathematical understanding, analytical reasoning, and problem-solving skills.

High School Mentor
Supported younger learners through mentorship and academic guidance.

Community Volunteer
Participated in community outreach activities assisting residents at a local old-age home.

Core Strengths

Problem Solving • Analytical Thinking • Teamwork • Communication • Attention to Detail

Interests

Software Development • Artificial Intelligence • Web Applications • Mathematics • Soccer